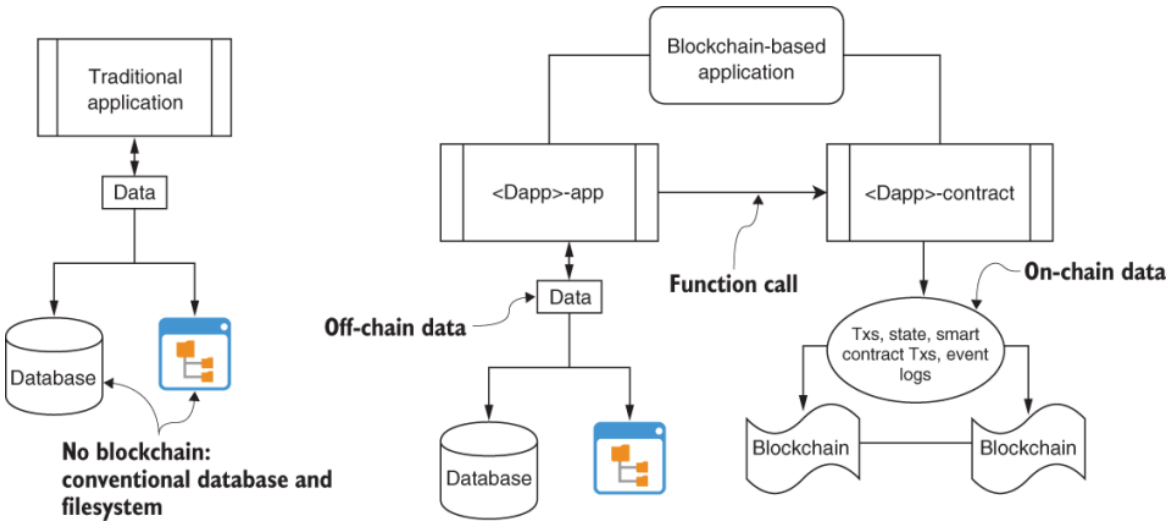


Figure 6.1 Traditional application vs. blockchain Dapp with on-chain and off-chain data

The left side of figure 6.1 shows a traditional application, with its conventional data stores. The right side of the figure shows the same system enhanced by blockchain recording of on-chain data. Thus, a blockchain-based system is a part of a larger system, which may be an enterprise system or a web-based system that stores data in a conventional database or on a local filesystem. An enterprise system, for example, may manage all its business data in a centralized private and secure database; it may also maintain a blockchain-based system for its decentralized operations. So in a typical business system, you'll have to deal with both categories of data: off-chain and on-chain.

> **NOTE** Understand that when you're developing a Dapp, you aren't porting your traditional system into the Solidity language; you're coding only the parts of a larger system that need blockchain support.

Given a decentralized scenario, a significant task in designing a Dapp is identifying the following:

- The activities that are the responsibility of the traditional part of the larger system